Transportation Research Part A 46 (2012) 1424–1448

Contents lists available at SciVerse ScienceDirect

Transportation Research Part A

journal homepage: www.elsevier.com/locate/tra

The private and social cost efficiency of port hinterland container distribution through a regional logistics system

Fedele Iannone *

Department of Economics and Quantitative Methods (DIEM), Faculty of Economics, University of Genoa, Via Vivaldi 5, 16126 Genoa, Italy

ARTICLE INFO

Article history:
Received 24 July 2011
Received in revised form 16 April 2012
Accepted 16 May 2012

Keywords:
Green freight logistics systems
Transport external costs
Modal shift policies
Port hinterland container connections
Interport model
Total social generalized logistic cost

ABSTRACT

Increasingly, the debate on freight transport and logistics involves the challenge of sustainable development. Key objectives of sustainable or "green" freight logistics systems are the mitigation of negative environmental and human health effects of distribution operations and the realization of a major modal shift in transport preferences, while at the same time achieving internal generalized cost efficiency and quality of services. Pursuing these goals requires the introduction of a range of measures. These measures call for private and public actors to take up various initiatives and adopt policies. Usually, it is more effective to combine different actions into an integrated package of measures than to introduce single instruments in isolation.

This article explores the nexus between sustainability and port hinterland container logistics. In particular, the methodology and results of an empirical analysis based on applications of a network programming tool called the "interport model" are presented and discussed. The model enables an examination of all possible effects on inland container flows and their associated internal and external costs due to public and private initiatives in the field of port hinterland container logistics. The empirical analysis aims to evaluate the impact of a set of simultaneous policy options and operational measures on the competitiveness and sustainability of hinterland multimodal distribution of import and export containers handled at the seaports of the Campania region located in Southern Italy. The loading units can transit through the dry port facilities (the so called "interports") located in the same region and/or through extra regional railway terminals, before reaching their ultimate inland destinations or the seaports. The integrated package of measures simulated by means of the model includes: (i) infrastructure policy, (ii) improvements of rail services, (iii) regulatory changes in terms of customs authorizations and procedures, (iv) removal of technical and legal barriers to fair and non-discriminatory competition in the market of rail traction between regional seaports and interports, (v) new business models integrating container logistics operations between seaports and interports, and (vi) social marginal cost charging of transport operations. Once this package of instruments is introduced, higher private and social cost efficiency of port hinterland container distribution through the investigated regional logistics system can be achieved. For instance, it has been estimated an annual saving of the order of about 12,660 tonnes of CO_2 equivalent emissions from transport corresponding to an external cost reduction of 0.27 million euros from the observed real life situation, whereas the estimated saving in terms of air pollution (CO, NO_x, PM, SO_2, VOC) from transport is approximately 220 tonnes per year corresponding to an external benefit of 1.31 million euros.

* Tel.: +39 (0)10 209 52 35; fax: +39 (0)10 247 55 11.
E-mail address: f.iannone@email.it

0965-8564/$ - see front matter © 2012 Elsevier Ltd. All rights reserved.
http://dx.doi.org/10.1016/j.tra.2012.05.019

The most immediate priority appears to be the customs and intermodal logistics integration of seaports and interports by means of full implementation of the "extended gateway" concept as a way to increase the rail share of modal split and improve the overall cost efficiency of the system. In addition, the simultaneous introduction of a social marginal cost charging policy can contribute to make the regional interports a viable solution to expand the hinterland reach of the regional seaport cluster.

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1. Introduction

1.1. Green logistics and transport external costs

The transportation policy, planning and research communities are moving towards the paradigm of sustainable or "green" logistics, which implies a broader perspective to address problems and find solutions in the freight sector (see for instance: European Commission, 2007; McKinnon et al., 2010; Rodrigue et al., 2001; Thore, 1991).

The approach towards the distribution of goods has greatly evolved in recent decades, firstly recognizing the need to see transport as a fundamental element of larger logistic strategies of businesses. Moreover, particularly within the last decade, the concerns related to the challenge of sustainable development have become increasingly important in freight logistics, with issues related to transport mode share and the internalization of transport external costs featuring strongly.

Sustainable or green freight logistics systems are concerned with the movement of goods, taking account of the negative environmental and social impacts of operations. Thus the objectives pertain not only to private aspects, such as cost minimization or profit maximization, but also to the wider effects of logistics on social well being, mainly in terms of the negative externalities – also referred to as "external costs" – associated with climate change, pollution, congestion, accidents and other biological and ecosystem damages deriving from transport. The challenge lies in minimizing these impacts while achieving strong logistic benefits in terms of total internal generalized cost reduction of distribution operations and service level improvement.

In any sector there cannot be economic efficiency unless prices reflect all costs to society actually caused by economic agents. The costs of transport can be split into private or internal costs and external costs. The former are directly borne by the users of transport system, while the latter, such as those indicated above, are imposed on others but not supported by the users. Transport external costs are a form of market failure and, without policy intervention, they do not directly affect the decisions about mode, route and timing of movements. The sum of private and external costs gives the social cost of transport (see for instance: European Commission, 2008a, 2008b; Nash, 2003; Pigou, 1920; Prud'homme, 2001; Rothengatter, 2003; Verhoef, 1996; Yang and Huang, 1998).

Negative externalities caused by transportation are widely recognized as a major problem in Europe and worlwide. Road transport is responsible for by far the largest share in transport external costs. Accordingly, a modal shift from unimodal road transport to less impacting modes is an accepted policy goal at various institutional and geographical levels. In this respect, much attention is currently being given to intermodal transport solutions particularly in inland freight sector. Intermodality today generally refers to the movement of cargoes in standardized loading units, using two or more modes of transportation. The partial or full transfer of containers, swap bodies and semi-trailers from road to rail or waterway mitigates the negative external impacts of transport operations, thus enabling the pursuing of efficient and greener freight logistics systems. The European Commission has also introduced the concept of "co-modality" to define a new policy approach promoting integrated transport systems where single modes do not necessarily compete against each other but rather complement one another, in the aim to obtain an efficient and sustainable utilization of resources (European Commission, 2006, 2007).

1.2. Private initiatives and public policies for sustainability in freight logistics

The achievement of private and external cost efficiency in freight logistics requires simultaneous actions by public and private actors as well as a close cooperation among stakeholders in order to create synergies.

Private companies can support the greening of logistics by adopting new technologies and/or by introducing new business models involving innovation in operations. Some of them are even starting to implement sustainable approaches to logistics and supply chain management as part of their Corporate Social Responsibility (CSR) agenda (see for instance: Sarkis, 2006).

Although logistic activities are mainly carried out by private operators, governments are still responsible for influencing the environmental and human health effects of cargo distribution operations. They have at disposal a range of policy measures to tackle sustainability problems in freight logistics, reducing transport external costs and inducing modal shift. Such measures can affect the demand or supply of transport and, in general, are not mutually exclusive. Their successful implementation may also require a complex interplay of different political levels at local, national and supranational scale.

Market based instruments rely on price mechanisms directly or indirectly addressing the so called "internalization" of transport external costs by means of financial penalties or incentives such as, for instance, charges, taxes and modal shift grants. Via monetary inducements, the users of transport systems are encouraged to reduce the negative external impacts they generate. More specifically, the internalization of external costs means making such effects part of the decision making process of transport users and it is therefore a policy instrument to ensure fair and efficient pricing within and between transport modes, while also altering the relative competitiveness of modes and favoring more environmental and social friendly logistic chains. Internalization eliminates the market imperfections due to externalities and defines a new equilibrium in which individual behaviours are consistent with the maximization of social welfare.

The level of economic measures aimed at internalization should ideally be based on the so called "social marginal cost charging" approach (see for instance: European Commission, 2008a; Nash, 2003; Prud'homme, 2001; Rothengatter, 2003; Yang and Huang, 1998). Transport prices expressed in kilometer charges should therefore correspond to the additional short term social cost created by one extra user of the infrastructure. However, within most countries, governments are still a long way from full application of the social marginal cost charging to the freight sector, as well as from reversing the long term modal shift to road. On the other hand, the benefits from internalization can only materialize when, for reason of fairness, not only freight movements but all transport sectors and all modes are subject to correct pricing of external effects, at least in a long term perspective. The internalization of the environmental and health costs of freight transport is currently high on the transport policy agenda in the European Union and is likely to be implemented over the next 10 years.

Furthermore, social marginal cost charging will only be effective insofar as transport choices are sensitive to price variations. In the presence of rigidities resulting from, or combined with different market failures (e.g. lack of alternatives, bounded rationality, etc.) that make demand less responsive to economic measures for internalization, other policy instruments can be used in mutually reinforcing ways to create greater price elasticity of demand and to make the supply of certain services more attractive (European Commission, 2008a, 2008b, 2011).

The presence of multiple market failures in transport sector suggests the adoption of a combination of individual instruments that complement each other and create a comprehensive policy mix. Besides market based measures, the most important government policy tools to achieve sustainability in freight logistics are: (i) regulation (including, amongst others, competition policy, administrative and technical simplification and harmonisation policy, working time directives, vehicle access restrictions, and the setting of standards), (ii) infrastructure policy (including the development of new infrastructure systems as well as the maintenance, repair, upgrading and optimal use of existing networks), (iii) land use planning (also including the siting of freight terminals, logistic platforms, etc.), (iv) innovation policy (also including the adoption of new technologies of information and communication), (v) research policy, and (vi) marketing and information policy.

There is not a single optimum package of instruments for all territories. The proper policy mix will always require to be tailored to specific circumstances such as geography, infrastructure development, freight logistics market, industrial strategy, and the weighting attached by policy makers and the public to different negative external effects from transport. In any case, the bundling of different policy measures can lead to synergies with the introduction of one type of instrument reinforcing the impact, and offsetting the disadvantages, of another.

1.3. Port hinterland container logistics and the extended gateway concept

The landside distribution of maritime containers from and to seaports is a complex process that affects the internal costs and the negative external impacts of production–distribution and logistic supply chains, having implications in terms of private and public decision making.

The actual modes of transport for port hinterland container flows are road, rail and barge. Transport organization can be done either by production and distribution companies or by their third party logistics providers in the case of "merchant haulage". Another possibility provides the arrangement of inland container transport directly by shipping lines and marine terminal companies in the cases of "carrier haulage" and "terminal operator haulage", respectively. In addition to pure transport operations, the port hinterland logistics of containers also involves a range of activities at seaports and inland facilities such as for instance the handling and storage of loading units, and the activities related to various types of clearance and inspection procedures.

The hinterland forwarding of containers has received a great deal of attention lately due to issues concerning the continuous growth of international trade, the introduction of bigger ships into the main trade lanes, and the possibility to raise the competitiveness of seaports, territories and productive systems through the adoption of innovative practices in inland intermodal distribution.

The steady increase in international container traffic volumes worldwide is putting great pressure on maritime terminals and port hinterland networks, particularly road networks surrounding the major port cities. For instance, approximately a quarter of the annual total inland freight transport demand within the European Union is port related and is mainly met by road (NEA, 2010), thus leading to undesirable levels of negative effects on environment and society. A crucial challenge many containerports are currently facing is the congestion due to insufficient storage space, long dwell times[1] and poor in-

[1] Dwell time is the length of time a container remains at a terminal before being loaded onto a transportation vehicle (ship, train, truck, barge) for further distribution. It is a critical factor correlated to terminal capacity. In particular, it is affected by: (i) customs and other administrative control procedures; (ii) the terminal operator's service level; (iii) the shipper's supply chain management strategies.

land transport. All these criticalities generate a need for new port hinterland distribution solutions essentially based on the combination of different transport modes and multimodal interchange nodes.

Sustainable port hinterland container logistics implies an efficient and environmental and social friendly landside distribution system supporting the container traffic of seaports. This puts emphasis on the use of rail and barge intermodal transport rather than unimodal truck transport. In addition, the development of inland multimodal nodes according to the so called "extended gateway" concept is a relevant option to reduce congestion phenomena and to improve the accessibility and connectivity of port hinterland container logistics systems.

The multimodal interchange nodes in the hinterland network of containerports are described differently in different countries, such as for example "inland ports" or "inland terminals" in the United States and Canada, "strategic rail freight interchanges" in the United Kingdom, "dry ports" in Sweden and other European countries, and "interports" in Italy as an abbreviation of "interior ports" (see for instance: Cullinane and Wilmsmeier, 2011; Harrison, 2007; Hayut, 1980; Iannone et al., 2007; Iannone and Thore, 2010; Jaržemskis and Vasiliauskas, 2007; Kirkland, 2007; Leitner and Harrison, 2001; Rodrigue and Notteboom, 2009; Rodrigue et al., 2010; Roso, 2008; Roso and Lumsden, 2010; Thore, 2007; UNCTAD, 1982, 1991). These facilities can contribute to the moving of traffic off the roads and onto rail and/or inland waterways, and to the improvement of cost and service efficiency of supply chains, enabling green and integrated freight logistics operations and stimulating regional economic growth.

The extended gateway concept incorporates the idea that some seaport facilities and functions can be duplicated and/or complemented at hinterland locations such as the interports (Franc and Van der Horst, 2010; Iannone, 2012; Iannone and Thore, 2010; Rodrigue and Notteboom, 2009; Veenstra et al., 2012; Visser et al., 2007). In particular, the availability of authorized areas dedicated to customs controls and other administrative and inspection procedures for the clearance of international freight shipments can qualify interports as an integral part (that is an extension) of specific seaports. Under a regime of customs continuity, the containers can be transported between seaports and interports without the need for inland customs transit documentation. This trade facilitation permits to save a great deal of time for the technical and administrative operations of container release in seaports and is the basis for sustainable transport.[2] Shippers and/or their logistics agents can arrange directly in the interports all the formalities in relation to the clearance of cargoes, whereas the organization of transport between seaports and interports is done by shipping lines and/or marine terminal companies. Moreover, governments can increase their revenue from taxes due to the generally acknowledged positive link between trade facilitations and freight flows.

The implementation of the extended gateway concept within a regional logistics system requires cooperation and integration among different actors, including infrastructure managers, shipping lines, inland intermodal carriers, customs, and so on. Hence, there are a number of challenges behind the development of interports according to the extended gateway perspective: new partnerships have to be established, new business models have to been developed, and transparency of goods and information flows has to be achieved.[3]

1.4. Plan for this research

Motivated by the relevance of the topics discussed so far, this article investigates a range of conditions to achieve private and social cost efficiency in the inland multimodal distribution of maritime containers imported and exported in Italy through the seaports of Naples and Salerno located in the Campania region in south of the country. The standardized loading units discharged and to be loaded in these seaports can transit through the interports of Nola and Marcianise located in the same region, as well as through extra regional rail intermodal terminals, before reaching their inland and port destinations.

The analysis of the existing situation concerning the port hinterland distribution of containers from and to Campania shows a general inefficiency and a number of other criticalities associated to the whole configuration and organization of the regional container logistics system. In order to precisely assess the optimization potential of such system, the impacts of a set of simultaneous policy instruments and operational measures aimed to increase the competitiveness and sustainability of port hinterland container logistics operations at the Campanian seaports and interports have been analyzed with the help of a mathematical programming tool called the "interport model". This analytic instrument incorporates economic, environmental and social parameters of inland container operations. It solves for the cost optimal flows of loading units throughout a port hinterland network featuring: (i) road connections, (ii) rail connections (with capacity constraints), (iii) regional interports and extra-regional locations served by road and railway, and (iv) regional and extra-regional locations served only by road. Differently from the conventional four steps models of transport demand forecasting (generation, distribution, modal split and assignment), the *interport model* combines the modal split and the assignment in a single step, allowing the simultaneous choice of transport mode and route including transhipment points.

[2] For instance, in compliance with the customs regulations currently in force in Italy, only the railway transport permits the necessary conditions of fiscal safety related to the inland haulage without any accompanying inland transit document for containerized cargoes that have not been nationalized yet through customs clearance.

[3] A successful example of extended gateway logistics is represented by the system of terminals and services arranged by Europe Container Terminals (ECT) to serve the hinterland traffic of containers transiting through the port of Rotterdam, in the Netherlands. ECT has introduced important innovations in port hinterland container distribution such as the terminal operator haulage practice, and the cross-border extended gateway concept at a trans-European level.

The major novelty of this capacitated network programming tool for port hinterland container logistics analysis and planning consists of the detailed modelling of container release operations and their associated pricing mechanisms at seaports and interports, including the possibility for shippers to relocate storage and customs operations from the seaports to the interports (i.e. the extended gateway concept). The model can simulate alternative long term logistic scenarios in terms of supply of infrastructure and services, demand characteristics, and government and industrial policies.

For the purpose of this work, two optimization scenarios have been formulated and solved by means of a high level computer programming language. These represent "ideal" policy scenarios simulating a range of public policies and private initiatives to improve the inland container operations at the Campanian regional logistics system. In detail, the public policy measures and the changes to operating practices that have been evaluated by means of the *interport model* are: (i) reactivation and/or improvement of a number of rail container connections from and to the regional seaports and interports, (ii) attribution of a fully operational customs status to the regional interports, so that these can be used as extended gateways where technical and administrative operations for the release of containers transiting through the regional seaport system can take place; (iii) internalization of the external costs deriving from inland transport flows.

The point (i) implies changes both in infrastructure policy and in provision of services. The point (ii) involves regulatory changes in terms of customs authorizations and procedures; in addition, it also implies some kind of vertical integration between seaports and interports as well as the facilitation to private companies in access to the market of rail traction services between seaports and interports. The point (iii) is assumed to indirectly represent a government scheme for funding rail container transport by means of incentives calculated as the total external cost savings arising from shifting containers from unimodal road transport.

The two ideal scenarios simulated by the model assume the introduction of the same public and private measures except one. In particular, a first modelled scenario, inspired by the work of Iannone (2012), does not take account of transport externalities in its objective function, while the second one internalizes in the objective function the transport external costs deriving from climate change, air pollution, noise, accidents and congestion.

The results in terms of logistic flows and total internal, external and social costs deriving from the optimal solutions of these scenarios have been compared with the observed real life situation. The external diseconomies of transport flows have therefore been calculated also with respect to both the existing situation and the modelled scenario whose objective function does not include the externalities. In addition, the comparison between real situation and ideal scenarios have been also made in terms of atmospheric emissions arising from inland transport operations, but the parameters representing such physical impacts have not been incorporated into the structure of the empirical models. Finally, a sensitivity test has been carried out to assess the modal split effects due to changes in the level of road transport fares. The results attained from this comprehensive analysis can constitute a useful knowledge base for regional initiatives aimed at stimulating new port hinterland container logistics solutions addressing modal rebalance and the pursuit of social welfare.

The rest of the paper proceeds as follows. Section 2 contains a review of literature contributions dealing with green logistics and the external costs of freight transport, with a focus on port hinterland container logistics research and the original features of the model-based empirical analysis presented in this article. Section 3 provides a conceptual description of the *interport model*, including its early formulations as well as its assumptions and limitations. Section 4 presents the sea-land container logistics system of the Campania region and illustrates and discusses the general features and main results of the applications of the *interport model* to the inland distribution of import and export containers through the Campanian seaports and interports. Section 4 addresses conclusions and possible future research developments.

2. Literature review

As employed in this article, the concept of sustainable logistics emphasizes the necessity of reducing transport external costs along supply chains, as well as the issues associated to the identification of appropriate public policies and private actions to achieve this goal, stimulating a shift of cargo movements to environmental and social friendly transport solutions. These topics are particularly relevant, amongst others, in port hinterland container logistics.

A critical review of the development of green logistics as a novel field of study is provided in the work of McKinnon (2010a), who examined the development of the subject over the past 50 years and proposed an analytical framework for investigating green logistics issues. Instead, a review of policies that can be used to rebalance the freight modal split is provided by Ricci and Black (2005) and McKinnon (2010b). The former authors focused on regulatory measures and economic instruments affecting intermodal transport, while also measuring the marginal social costs of road and intermodal transport on specific corridors covering a variety of European countries. The latter author identified and discussed the following six categories of policy: (i) taxation, (ii) financial incentives, (iii) regulation, (iv) liberalization, (v) infrastructure and land-use planning, and (vi) advice and exhortation.

The negative external impacts of transport and their associated costs have been the subject of extensive research during the past years. A recent review of the different external cost categories and their associated estimation methods is provided in the handbook by Maibach et al. (2008) produced within the IMPACT project (Internalization Measures and Policies for All external Cost of Transport) funded by the European Commission.

A number of empirical studies in scientific literature have identified clear benefits of rail and water movements over road haulage and have emphasized the potential role of charging the external costs in contributing to modal shift. Forkenbrock

(1999) estimated the external costs arising from intercity freight transport by truck in rural areas of the United States of America. The external costs considered are accidents, atmospheric emissions, noise pollution and unrecovered costs associated with the provision, operation, and maintenance of public facilities. Such costs were also compared with private operating costs. Forkenbrock (2001) extended his earlier work on truck transportation to include rail transportation and made a comparison between the two modes. The external costs covered in this analysis are those associated to accidents, air pollution, climate change, and noise.

Beuthe et al. (2002) assessed the external costs of pollution, congestion, accidents, noise and road damages arising from road, rail and inland waterway traffic in Belgium. The authors used a GIS-based optimization modeling methodology and also estimated the impacts on modal choice and in terms of external benefits deriving from the internalization of external costs into the users' total costs.

Feng and Huang (2005) formulated a multiobjective mathematical programming model optimizing multimodal intercity logistics flows. The model solves for both the least cost routing of shipments and the optimal locations of railway terminals. Total costs in the model also include the transport externalities associated to atmospheric pollution and accidents. The model was applied to the Taiwan area.

Janic (2007) developed a calculation model for analyzing the full costs of operating intermodal and road transport networks. Full costs are the sum of internal and external costs, where external costs consist of costs from air emissions, congestion, noise pollution and accidents. The model was applied to simplified network configurations in Europe in order to assess the freight modal competition effects arising from internalization.

Recently, a number of studies carried out by international institutions have theoretically addressed the new role of port hinterland container connections in trade logistics, with consequent economic, environmental and social dimensions (see for instance: European Commission, 2009a; OECD, 2012; OECD and ITF, 2009; UNECE, 2010). The simultaneous consideration of all these dimensions is a critical element for promoting and attaining sustainable inland container logistics systems. Nevertheless, the incorporation of the negative external costs of transport into the analysis and planning of container distribution from and to the hinterland of seaports is still relatively little explored and rarely quoted in scientific literature.

Berechman (2009) estimated the full marginal costs associated with the expected additional truck traffic resulting from expanding trade activities at the Port of New York and New Jersey. Full marginal costs include private operating costs, infrastructure costs, and the external costs of congestion, safety and air pollution. Chang et al. (2010) analyzed the potential to shift container cargoes from road to short sea shipping (SSS) and train in Korea by means of an optimization model minimizing internal transportation costs and the external costs from air pollutants and greenhouse gases. Lee et al. (2010) developed a calculation model to compare the external costs of air pollution and climate change between truck and SSS on a corridor basis for domestic container transportation in Taiwan. Liao et al. (2011) calculated the total carbon dioxide (CO_2) emissions from inland container transport in Taiwan during the time period 1998–2008, and used a regression model to estimate the trend, identifying the determinants of such emissions. Macharis et al. (2010) employed a GIS-based optimization model to assess how the market area of the intermodal terminals in Belgium could vary in response both to fuel price increases and to a policy measure of internalization of transport external costs. Various comparisons were conducted in order to find shortest paths and the associated transport costs for container flows from the port of Antwerp to each Belgian municipality via intermodal terminals and via unimodal truck transport.

The empirical analysis presented in the present article innovates in the context of port hinterland container logistics research under different aspects. For instance, none of the studies reviewed above has focused on areas in the Mediterranean Europe. Moreover, none of these studies has evaluated the benefits arising from including the internalization of transport external costs in a wider set of measures addressing the rebalance of modal split.

This article intends to contribute to the literature by analyzing a sustainable port hinterland container logistics problem in Italy. The impacts of simultaneous policy instruments and operational measures to improve the competitiveness and sustainability of inland container operations at the sea-land logistics system of the Campania region have been evaluated by means of a novel modelling tool called the *interport model* and whose objective function incorporates also the external costs associated to climate change, noise and air pollution, accidents and congestion arising from inland transport flows.

For several years, the Campanian policy makers have been calling for more and more freight traffic to be shifted from road to rail. They have also been advocating a more active role of the interports in relieving the congestion phenomena induced by container traffic, especially in the port of Naples and over the road transportation system of the region. The ongoing trends and future challenges point the need for satisfying a rising demand for accessibility and connectivity in a context of growing sustainability concerns. Nevertheless, the situation is still far from showing any real sign of improvement. In such a context, more effective policy formulations require an in depth assessment of the current and potential performances of the regional container logistics system, and more specifically a detailed knowledge of the internal and external costs of inland container distribution and of the driving factors behind them. At present, there are many uncertainties about future policy changes and their impacts. To better cope with these uncertainties, a model-based analysis is needed to inform decision makers about possible container flows and their associated social costs under different scenarios.

To the best of our current knowledge, the *interport model* is the first mathematical programming tool for the economic analysis and strategic planning of port hinterland container logistics which takes into consideration customs clearance issues and the extended gateway concept. It allows simulating the possibility of different types of customs control on full containers at seaport and interport nodes, as well as the possibility of relocating at the interports the operations of final technical and administrative release of containers that are usually carried out in the seaports.

Similarly to the other model-based analyses reviewed above, also this research was not intended to estimate the marginal value of either cited different external impact of transport, so that recently published parameters valid for Italy have been used as basis for monetary and physical calculations of the transport external effects associated to the port hinterland distribution of containers from and to the Campania region.

3. Methodology

This section presents in general conceptual terms the network programming model employed for carrying out the empirical analysis illustrated in the next section. It also includes a comprehensive discussion about the assumptions and current limitations of the model.

A complete illustration of the mathematical formulation of the model is beyond the scope of the main text of the present article. On the other hand, the structure of the model essentially represents a novel extension of that characterizing the *import/export simultaneous interport model* proposed by Iannone (2012). Unlike its earlier formulation, the model now also incorporates a wide range of external cost parameters associated to inland transport operations.

Such analytical tool allows measuring the economic, environmental and social benefits arising both from shifting the seaport exit/entry gate of import/export containers to interports (the extended gateway concept), and from employing intermodal transport for port hinterland container distribution. More generally, it enables an examination of all possible impacts on flows and their associated internal and external costs due to public and private initiatives in the field of port hinterland container logistics.

3.1. The interport model

The methodology employed for the empirical analysis presented in this article builds on that of earlier preliminary researches carried out by Iannone and Thore (2010) and Iannone (2012).

Iannone and Thore (2010) formulated the *inward interport model* as a combined mode choice/route choice modelling tool optimizing the port hinterland multimodal flows of imported full and empty containers under the perspective of the controllers of cargoes (merchants and shipping lines).[4] The main purpose of the model is to highlight and measure possible advantages arising both from extending the seaport exit gate of import containers to interports, and from employing intermodal solutions for port hinterland distribution.

The model features linear parameter and constraints. The primal objective function[5] to be minimized represents the total internal generalized cost of inland container logistics operations throughout a distribution network composed of regional seaports and interports served by road and railway, extra regional locations served by road and railway, and other regional and extra regional locations served only by road. The model also incorporates a road supply sub-model for the quantification of transport time on road links according to the Road Code regulations.[6] As for the total travel times by rail over admitted connections, these are instead exogenous.

The critical direct and indirect internal cost items explicitly taken into account by the *interport model* are: (i) container handling costs, (ii) container storage costs at seaports and interports, (iii) costs for physical inspection and X-ray scanner control by customs at seaports and interports, (iv) in transit inventory holding costs, (v) container leasing costs, and (vi) container transport costs. The primal optimization is subjected to flow balance constraints at all nodes, non-negativity constraints on endogenous variables, and capacity constraints on all rail connections. Flow balance and non-negativity conditions represent conventional constraints in network programming models.[7]

Multimodal freight models are traditionally built following the well known four steps modelling methodology in which generation, distribution, modal split and assignment are seen as separated modules. In opposition, the primal program of the *interport model* solves simultaneously for the optimal modal and nodal split of inland traffics of containers transiting through a regional seaport system.

[4] The authors applied the *interport model* to optimize the inland distribution of containers imported in Italy through the seaports of the Campania region. The results obtained from the optimization of three different scenarios simulated by the model were compared to the observed real life situation in 2007. One of these three modelled scenarios was created to reproduce the existing situation and therefore to validate the model.

[5] Optimization models can be formulated in either minimizing or maximizing form. In either case, in linear programming the originally formulated model is called the "primal program". For every primal program there is an implicitly related "dual program" which involves the same data, provides useful information for sensitivity analyses, enhances the understanding of the original model, and allows increased insights into the interpretation of problem solution. Each constraint in the primal problem has an associated shadow price or dual variable which indicate how much one would be willing to pay for additional units of given limited resources, thus representing the marginal utility of relaxing the constraint or equivalently the marginal cost of strengthening the constraint. Apart from the constraints, also all variables of a primal programming problem have what is known as a marginal or imputed value. Compared with other programming model types, linear models are by far the easiest to solve. Optimal values of primal and dual variables are given by modern computer programs as part of the optimal solution.

[6] In empirical applications of the model, total travel times on admitted road links are computed as the sum of driving time and the time for rests and stops prescribed by Italian Road regulations under the "one driver on board" hypothesis. Driving times are computed by assuming two different admitted truck's average speeds on motorways and other road types. The number and time duration of rests and stops are calculated as a function of driving time.

[7] Flow conservation at origin nodes requires that the supply at each node must suffice to cover the flows leaving the same node. For intermediate nodes the balancing conditions state that the flow entering each node must suffice to cover the flows leaving it. For each destination node, the deliveries forthcoming at the node must suffice to cover demand. Finally, non-negativity constraints state that the primal variables cannot assume negative values.

Within the model, a single interport infrastructure is represented by a pair of "virtual" nodes that are supposed to have the same geographical location but involve in part different interport processing activities. One virtual node has a local demand of containers, while the other virtual node is essentially a pure intermediate node with customs function.[8] The splitting of a single interport facility into two separate nodes can be seen as a mathematical artifact of graphic theory enabling an easy modelling of the extended gateway concept. In fact, it allows formulating a standard linear programming model for the entire network, thus avoiding explicit 0–1 programming features to handle the decisions of where to carry out the customs and storage operations for full containers.[9]

As for the whole distribution network configuration in the model, seaports are supply or origin nodes, whereas the remaining nodes, excluding the virtual interport nodes with customs function, are ultimate destination or demand nodes. All inland locations served by railway, including interports, can be also used as intermediate nodes to perform multimodal transhipment operations along container traffic relations from origins to final destinations.[10]

Solving the primal program of the *inward interport model* includes finding the detailed quantities of containers to be shipped from the regional seaports, the transportation modal choice along each inland link, and the detailed pathway through the system chosen, including the possibility of container transhipment operations at the regional interports and at the other inland nodes featuring a railway terminal. In addition, the model also determines whether shippers will choose to have their containerized consignments controlled and cleared by customs directly at the seaports, or whether they will prefer to comply with customs formalities at the interports. The primal equilibrium solution can be therefore broken down into merchant flows and carrier flows, representing in any case the optimum of a hypothetical shipper operating the entire network.

By means of parameters representing dwell times, free of charge storage times, handling charges, demurrage charges, probabilities and costs of customs controls, the model simulates in detail the container release operations and their associated pricing mechanisms at seaport and interport terminals, including the possibility of relocating storage and customs operations from the seaports to the interports according to the extended gateway concept.[11] In this respect, the model also allows for spelling out various arrangements of customs checks on full containers (automated computerized controls, documentary control, X-ray scanning controls, and physical inspections). Of course, empty containers do not require customs clearance before being released from intermodal nodes.

Iannone (2012) expanded the initial formulation of the *interport model* by proposing a new model structure made up by two main components or primal subproblems. A first component, inspired by the work of Iannone and Thore (2010), minimizes the total internal generalized logistic cost associated to the port hinterland multimodal distribution flows of import containers (*inward* submodel), while the second one optimizes the distribution flows of export containers based on the same minimizing criterion (*outward* submodel). The primal program simultaneously optimizes the *import* and *export* submodels (*import/export simultaneous interport problem*) subjected to the same type of constraints of the early model of Iannone and

[8] A first node of the pair of virtual nodes representing a single real interport infrastructure is assumed to be a demand node performing only multimodal switching and logistic warehousing functions; also, it is supposed to be connected to specific seaports by road and/or rail links and is assumed to be employed only for the release operations (handling, storage and forwarding) concerning both full containers already cleared by customs at the seaports and empty containers. The second virtual node of the pair is supposed to be a pure transhipment node performing only the functions associated to multimodal switching and customs clearance and inspections of full containers transiting through specific seaports; hence, it does not feature a local container demand and is assumed to be employed only for the release operations (handling, storage, customs checks and forwarding) concerning the full containers not cleared by customs at the seaports. Based on the customs regulations currently in force in Italy, such virtual node is supposed to be connected to seaports only by railway and exclusively for the transfer of full containers under the responsibility of shipping lines (carrier haulage) and without any accompanying inland customs transit document. Both virtual nodes feature the same outbound road and rail connections. In addition, road transport at a nil generalized cost is admitted between the two virtual nodes to meet the demand expressed by importing operators located in the interport for full containers to be cleared by customs at the same interport. Finally, each railway service available at a single real interport is represented simultaneously by the specific rail connections available at each corresponding virtual node. Such connections are tied together in the capacity constraint of the corresponding single service.

[9] In multimodal freight transport modelling, the representation of simple geographic networks does not always provide an adequate basis for detailed analyses of different distribution operations that can be performed on single infrastructures. For instance, freight terminals can be used in different ways and based on different time and cost parameters. To simulate the different container processing activities that can be performed at a single interport facility, the basic idea in the *interport model* is to create two virtual nodes corresponding to the single real interport, with specific time and cost parameters for various particular activities. This approach generates much larger networks than their real geographic representation. A somewhat similar framework is also provided by the concept of "supernetworks" proposed by Sheffy (1985) and by the "virtual network" methodology introduced by Harker (1987) and further expanded in different ways by Crainic et al. (1990) and Jourquin and Beuthe (1996). For instance, the virtual network approach implemented in the NODUS software (Jorquin and Beuthe, 1996) proposes a methodology and an algorithm that creates in a systematic and automatic way a complete virtual network with virtual links corresponding to the different operations which are feasible on every real link or node of a geographic multimodal network. Other studies based on this methodology are those carried out by Beuthe et al. (2001), Beuthe et al. (2002), Geerts and Jourquin (2001), Jonkeren et al. (2011), Macharis et al. (2008).

[10] At the extra-regional nodes served by railway it is assumed the possibility to perform only rail-to-truck and truck-to-truck transhipment operations, while at the interports it is also assumed the possibility of railway-to-railway and truck-to-railway transhipment operations.

[11] Terminal operators offer container storage services to decouple the successive steps in the transport chain and they normally provide a limited amount of free of charge storage time which should allow for the customers to arrange customs formalities and other authorities to do their required clearances. Daily storage rates after free time (demurrage charges) are charged by terminal companies to optimize the yard productivity, minimizing the container dwell time depending on deliberate supply chain management strategies of using terminals as low cost warehouses. Also customs and administrative procedures may add substantially to the dwell time, and therefore to the direct and indirect costs for shippers. In some cases, shippers endeavour to have all necessary containers checked within free storage time provided by terminal companies, with this typically being not achieved for 100% of container volumes and determining significant generalized costs to be borne.

Thore (2010).[12] Within the new model structure, seaport nodes are at the same time origin nodes for the distribution of import containers (*inward* submodel) and destination nodes for the distribution of export containers (*outward* submodel). Extra-regional railway locations, virtual interport nodes without customs function, and locations served only by truck are simultaneously destination nodes in the *inward* submodel and origin nodes in the *outward* submodel. Virtual interport nodes with customs function are pure intermediate nodes because they do not feature a local demand and/or supply of containers. Finally, all the inland nodes served by railway can perform a multimodal transhipment function for various O/D combinations, that is, for traffic relations from supplying seaports to demanding inland locations (import), and from supplying inland locations to demanding seaports (export).[13]

As employed for the purpose of the empirical analysis illustrated in the next section, the *interport model* has been further expanded according to a sustainable logistics perspective. In particular, the model now internalizes in its objective function also the external costs of greenhouse gas emissions, air pollution, noise, accidents and congestion arising from road and rail container traffic. Ultimately, the *interport model* now simultaneously incorporates economic, environmental and social parameters of inland container distribution operations under a single objective framework. The primal programming problem minimizes the total social generalized cost of port hinterland container logistics subjected to flow balance conditions, non negativity of variables, and rail capacity limits. The total social logistic cost encompasses the total internal generalized logistic cost and the total transport external cost of port hinterland container distribution.

3.2. Assumptions and current limitations of the interport model

In treating allocations of containers from origin to destination points in terms of the *interport model*, a number of certain assumptions are made.

Firstly, the model supposes the rationality of the economic agents buying port hinterland container services and the presence of perfect competition in the market. In particular, the market simulated by the model is essentially composed of shipping lines and merchants from the demand side, and inland multimodal operators from the supply side. The representative optimizing agent of the model is a generic shipper selecting its own multimodal distribution choices incurring the minimum total cost. In the common fashion, the overall solution for this economic agent can be shown to coincide with the decentralized solutions of individual programs for each participating agent who needs to ship containers throughout the network. In these respect, shipping lines and merchants buying services in the market of multimodal operators are treated as "irrelevant intermediate alternatives".

Dual constraints and the complementary slackness conditions of the *interport model* impose "zero profit" conditions for distribution operations, thus characterizing a competitive, constant-return-to-scale port hinterland container logistics industry in which free entry drives excess profits to zero for used transport links and no unprofitable activities are operated.[14] The assumption of perfect competition is appropriate and it does not limit the realism of the model. As a matter of fact, inland multimodal operators are essentially specialized intermediary firms such as, for instance, intermodal marketing companies. This market features: (i) low barriers to entry, (ii) low fixed costs, and (iii) a high level of competition among similar sized firms. Usually, these companies do not own physical assets but process information and act as third party agents organizing and coordinating container transport and logistics activities by multiple solutions on behalf of shipping lines and merchants. They hire inland transport and logistics companies owning physical assets and coordinate the use of their equipments for the execution of physical services.[15]

The high level of competitive intensity in the Italian market of inland multimodal operators is documented in a research report realized within the "Diomis" project (UIC et al., 2006). Further considerations on the competition levels in inland container distribution markets are contained in a research report by UNCTAD (2004).

[12] At the empirical level, the author compared the results obtained from a modelled ideal scenario to the observed real situation concerning the inland container distribution operations through the seaport-interport logistics system of the Campania region. The analysis of Iannone (2012) also took into account the external costs and physical air emissions from transport, but their related parameters were not included in the model's structure.

[13] At the extra-regional nodes served by railway it is assumed the possibility to perform only rail-to-truck and truck-to-truck transhipment operations over the *inward* direction and truck-to-rail and truck-to-truck transhipment operations over the *outward* direction. At the interports it is also assumed the possibility of performing rail-to-truck, truck-to-railway, truck-to-truck and rail-to-rail transhipment operations over both the *inward* and *outward* direction. Furthermore, the two virtual nodes corresponding to a single real interport facility feature the same outbound and inbound road and rail connections over the *inward* and *outward* direction, respectively.

[14] The primal and dual programs of a linear programming model are tied together according the so called "complementary slackness conditions" (Thompson and Thore, 1992). Complementary slackness is the relationship between slack and/or surplus variables in the primal problem and the opportunity costs in the dual. Slack and surplus variables respectively convert $\leqslant$ and $\geqslant$ inequalities to equalities. These quantities are zero for those constraints that are satisfied exactly. In this case, the corresponding opportunity costs are positive. Instead, dual prices of zero are associated to primal constraints with a positive slack. The dual program of the *interport model* maximizes the total net appreciation of container flows' shadow value cumulating in the network, while also including the shadow value of rail capacity constraints, subjected to the conditions that the value appreciation created along each and every link be exhausted by its costs. Shadow value appreciation is the difference between the shadow price of the shipment at destination and the shadow cost at origin. More specifically, optimality requires that along any positive road flow in the network, the increase in shadow value must be exactly exhausted by the total unit logistic cost of the shipment. Along any positive rail flow in the network, the imputed appreciation along the rail link plus the shadow price of the capacity limit must be exactly exhausted by the total unit logistic cost of the shipment. The hypothetical shipper will then break exactly even. No free value can arise. All the unused links must reflect shadow value appreciations that are less than the logistic cost of transfer.

[15] Anyhow, it is also worth mentioning that, in practice, there is sometimes much difficulty in drawing a clear limit among the various activities performed in transport and logistics markets, such as for instance in the case of firms providing simultaneously intermediary (immaterial) services and physical services.

Other fundamental assumption is that the problem is formulated as a linear programming model. The objective function and the constraints of the *interport model* contain the sum of individual terms that are proportional to the value of variables. Moreover, the model is static and deterministic. All model's elements, from container demands to endogenous variables, cost structure and capacity constraints are for one planning period (corresponding to an operational year in empirical applications), and are assumed to be constant and known with certainty.

These assumptions have been designed to keep the model at a basic level, while also recognizing the power and importance of a linear modelling formulation. Indeed, large scale problems can be easily formulated as linear programming models, so that they can also be efficiently solved in short time with state-of-art solvers. Of course, most real problems feature non linearities as well as random parameters with accompanying probability distributions. For instance, there might be economies of scale that allow one to reduce costs as volume of activities increase. In this case, piecewise linear functions can be used to represent economies of scale in logistic costs associated with concentrated flows. In addition, also factors associated to service level such as reliability and congestion certainly determine mode and route choice. In this case, nonlinear stochastic programming is a better option as a means for creating more accurate models, but at the expense of increasing the computational burden due to the necessity of using complicated solution algorithms. Furthermore, the dynamic modelling framework for modelling transport congestion phenomena by means of internal and external non linear performance functions depending on driving time, value of time and link capacity utilization (BPR, 1964; Sheffi, 1985; Maibach et al., 2008) could prove to be quite complex unless more comprehensive network models able to simulate all traffic flows (including passengers flows) are available. Nevertheless, the congestion factor is somewhat taken into account by the *interport model* in the following ways. The static modelling framework adopts a segmentation of average flow speeds on roads according to specific types of travelled link. In addition, all rail connections feature capacity constraints. Finally, dwell time parameters are supposed to also internalize the negative effects arising from possible inadequate levels of handling, storage and/or customs capacity at maritime and inland nodes.

A specific aspect that deserves attention is the modelling of reliability, including elements related to just in time deliveries, safety stocks and their associated costs. Shippers or consignees may not be interested in fast deliveries, and yet a delay with respect to an expected delivery time may produce a great damage, sometimes even a far greater damage than what should have been borne to obtain a timely delivery by using a more expensive mode of transport. When investigating such cases, the valuation fundamentally depends on the value of goods and the needs of the consignees, with effects on the so called "safety" or "buffer" inventories. As from business logistics literature, buffer inventories can be calculated based essentially on: (i) the customer service level, (ii) the uncertainty in shipment lead time, and (iii) the demand forecast error. Usually, it is not an easy task obtaining detailed data on stated or revealed preferences of the consignees. Hence, when specific data concerning are not already available or when it is not possible to carry out an *ad hoc* study, a more conventional approach based on the opportunity cost of frozen capital may be used, such as the approach used in the *interport model*. In practice, this method only requires an evaluation of the so called "in transit inventory costs" by calculating the interest matured on the value of the goods in transit in relation to the average time duration of distribution operations (i.e. terminal and transport operations).

Moreover, the main challenge in port hinterland container logistics modelling seems to be that of calculating the unit safety stock costs at destination in relation to the movements of full containers distributed simultaneously by different and alternative solutions featuring different lead times. In the currently available literature on container network optimization, safety stock costs are calculated only relating to the deliveries of total batches of containers by single distribution solutions (Jula and Leachman, 2011; Leachman, 2008); this means that each destination is supposed to be served, for instance, by a single transport solution only. In our opinion, a good starting point for more realistic approximations of safety inventories in transport and logistics modelling are the analytical results recently achieved by Boute and Van Mieghem (2011) and Combes (2011). These contributions show how to estimate the safety stock for a single good that is sourced from two locations ("dual sourcing") or by using simultaneously different transport solutions.

Further assumption of the *interport model* is that the port hinterland network configuration is typified by relatively small inland locations featuring a railway terminal (including interport locations). Similarly to the other nodes of the network, also the inland locations served by railway are assumed as centroid areas in which the traffic of demanded and supplied containers terminates and originates, respectively. Given this configuration, pre and post haulage activities at these locations are assumed to serve negligible (very short) distances. Hence, at the moment, such activities are omitted from the modelling entirely.

Finally, the negative environmental and social impacts of distribution operations are currently entered into the model only in monetary terms, but a number of them, such as greenhouse gas emissions and air pollution, can be incorporated into the model's structure also in physical terms. Multiobjective modelling techniques are particularly suitable for decision problems involving trade-offs among conflicting goals such as the simultaneous minimization of internal logistics costs and physical environmental impacts of port hinterland container distribution operations. In this case, multiple non-dominated or "Pareto-optimal" solutions of the model can be obtained, for instance, by applying the *augmented ε-constraint method* (*AUGMECON*), first introduced by Mavrotas (2009).

4. Analysis

This section presents the current features and policy orientations concerning the container seaport–interport logistics system of the Campania region in Italy. It also illustrates and discusses the general features and main results of the application of the *interport model* to the simulation of two ideal policy scenarios optimizing the inland distribution of import

and export containers through the Campanian container load centres, while also describing the topological and functional characteristics of the large scale real port hinterland network under consideration.

The output data in terms of modal split and total costs resulting from primal optimizations are compared with the observed real life situation. The modelled scenarios and the real situation are also compared in terms of air emissions from inland transport. Furthermore, optimal dual prices associated to rail capacity constraints under the two simulated ideal scenarios are shown and interpreted. Finally, the results of a sensitivity test calculating the changes of modal split proportions as a function of road transport fares' variations are presented.

4.1. The container logistics system in Campania: current features and policy orientations

Campania is a region located in Southern Italy. The network system of first-tier sea-land container load centres in Campania encompasses the Mediterranean seaports of Naples and Salerno, and the interports of Nola and Marcianise (Fig. 1).

Naples and Salerno are import/export ports but with a low share of transhipment business as well. They essentially share the same hinterland, even if they serve and are served by different trade partners overseas (the so called "foreland"). The distance between these ports is just 50 km. Approximately 80% of the total import/export container traffic of the Southern Italy ports is handled at the Campanian seaport cluster (Iannone, 2011).

The intermodal facilities at Nola and Marcianise have been relatively recently constructed in order to both facilitate port container traffics and connect the region to long distance railway corridors. Both regional interports are located approximately 30 km from the port of Naples; furthermore, Nola is situated 57 km from the port of Salerno, while Marcianise 80 km. The distance between the two interports is just 32 km.

The main characteristics and difficulties of the Campanian regional logistics system for intermodal traffic and particularly for inland distribution of maritime containers may be summarized as follows: (i) port hinterland still exclusively limited to the national scale; (ii) low rates of utilization of the existing capacity of rail connections from and to seaports and interports; (iii) suspension of rail connections at the port of Salerno due to infrastructure safety purposes (since 2005); (iv) legal and technical barriers to fair and non-discriminatory competition in rail traction services between seaports and interports; (v) congestion on the road system throughout the region; (vi) port capacity saturation; (vii) severe congestion and high container dwell times at the port of Naples also deriving from slow customs procedures.

In the last years, much of the growth in container throughput and related congestion phenomena at the port of Naples has resulted from the rapid development of Chinese manufacturing industries (mainly textiles, clothing, footwear, and leather goods). Approximately 70% of the total containerized import traffic at Naples comes from China (Iannone, 2011). Naples is the first Italian port in terms of quantities imported from China but the last port in terms of customs declared values.

Just in 2010 the intermodal terminal company of the Marcianise interport was authorized to operate a customs bonded "A3 area" for handling, storage, inspection and clearance of international containers. Just in 2011 an X-ray scanner was

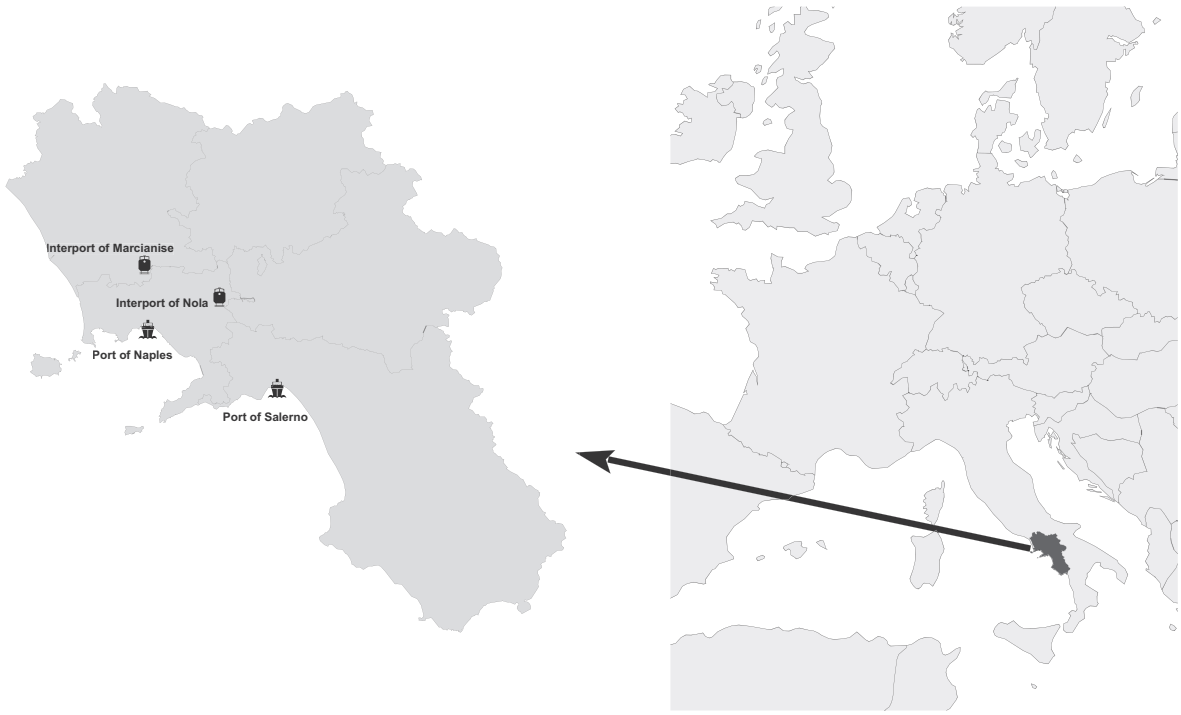


Fig. 1. The Campania region and its sea-land intermodal logistic system. *Source:* author's adaptation from Iannone and Thore (2010).

installed within this interport. Nevertheless, neither Nola nor Marcianise is currently employed as an extended gateway supporting the inland traffic of containers transiting through the seaports of Naples and Salerno. In Campania, some maritime terminal companies and freight forwarders are still reluctant to engage clear long term partnership with interports in the fear of core activities shift to other operators. In addition, there are not examples of shipping lines investing in interports.

On the other hand, by virtue of an ancient customs regime still in force in Italy, at the moment the rail connections between the Campanian seaports and interports can be employed only for merchant traffics if operated by private rail traction providers. Container forwarding solutions arranged by shipping lines according to the extended gateway concept (i.e. without the need for customs transit documentation) are possible only where rail services provided by the Italian State-owned rail carrier are available. "Trenitalia" is the incumbent rail traction provider belonging to the State-owned rail holding FS Group ("Gruppo Ferrovie dello Stato") and the sole company authorized to operate rail services under facilitated conditions for customs bonded containers travelling under steamship bills of lading from and to the Italian seaports.

At the end of 2007, for the first time, a number of containers were carried under an extended gateway regime between the port of Naples and the interport of Nola by Trenitalia under the responsibility of a major shipping line. The port dwell time for these containers was only 1.7 days. Unfortunately, since some year Trenitalia has ceased to operate rail container services between the Campanian seaports and interports due to severe financial problems that are bringing to a reorganization of the cargo division of the company. Possible future successful implementations of extended gateway systems in the Italian regions should therefore also imply greater liberalization and further types of regulatory change for rail traction service provision in port hinterland container markets.

A prospective solution identified by the Campanian policy makers to stimulate the use of interports and increase the modal share of rail container transport in the region would provide for a grant scheme for subsidizing container shuttle trains operated by a single new private rail traction company between the port of Naples and the interport of Nola (Regione Campania, 2007). In this respect, the European Commission has not raised objections to a hypothetical start-up state aid granted by the Campanian regional government (Aid No. 449/08 in European Commission, 2009b).

4.2. General features of empirical modelling applications to the Campania region

For the purpose of this work, two policy scenarios concerning the inland container operations that can be carried out during a year at the port–interport logistics system of the Campania region have been simulated by means of the *interport model* and based on real data. These represent "ideal" scenarios assuming the introduction of a package of simultaneous public policy instruments and operational measures aimed at increasing the internal and external efficiency of hinterland container distribution through the maritime and inland load centres of the region. In particular, both modelled scenarios assume a fully operational customs status of the Campanian interports. This has permitted to simulate the possibility for shipping lines to arrange the customs bonded railway transfer of full containers between seaports and interports according to the extended gateway concept. In addition, both ideal scenarios also suppose the availability of all container railway services operated at the Campanian regional logistics system during 2005–2007 (hence, also including the services operated at the port of Salerno). The annual two-way capacity assumed for each of these services corresponds to the maximum capacity surveyed from 2005 to 2007. Finally, the first modelled scenario, inspired by the work of Iannone (2012), does not incorporate transport external costs in its objective function. Indeed, this scenario minimizes the total internal generalized logistic cost of hinterland distribution of full and empty containers imported and exported through the Campanian seaport cluster. Conversely, the objective function of the second scenario features the internalization of the external costs associated to climate change, air and noise pollution, safety and congestion deriving from inland transport flows. Hence, in this case the model minimizes the social total generalized logistic cost of inland distribution of full and empty containers discharged and loaded at the regional seaports.

The empirical results in terms of logistic flows and total internal, external and social costs deriving from the modelling applications have been compared with the observed existing situation in 2007. The negative impact of transport activities have therefore been monetized also with regard to both the real situation and the modelled ideal scenario which does not incorporate the externalities into the objective function. The different scenarios have been also compared in terms of physical atmospheric emissions arising from transport flows, but the parameters corresponding to these impacts are not included in the modelling formulations. Finally, a post optimality test has been executed to assess the impacts of road transport prices' variations on modal split. The effects of these variations have been also compared with the effect of the internalization of transport external costs.

As in Iannone (2012), the real port hinterland container network under consideration features 24 nodes and 163 two way road and rail links (Fig. 2). It consists of the following seaports, interports and other inland locations in Italy:

- *Seaports*: Naples and Salerno.[16]
- *Interports*: Nola and Marcianise.[17]

[16] In the empirical applications of the model, the Campanian seaports are coded as NAP and SAL.

[17] Fig. 2 only represents real nodes and real links. Actually, the number of nodes and links actually included in the formulation of the empirical models is higher due to the representation of each real interport facility by means of two virtual nodes. The virtual nodes without customs function corresponding to the interport facilities at Nola and Marcianise are respectively coded as NOL and MAR, whereas their associated virtual nodes with customs function are coded as NCC and MCC.

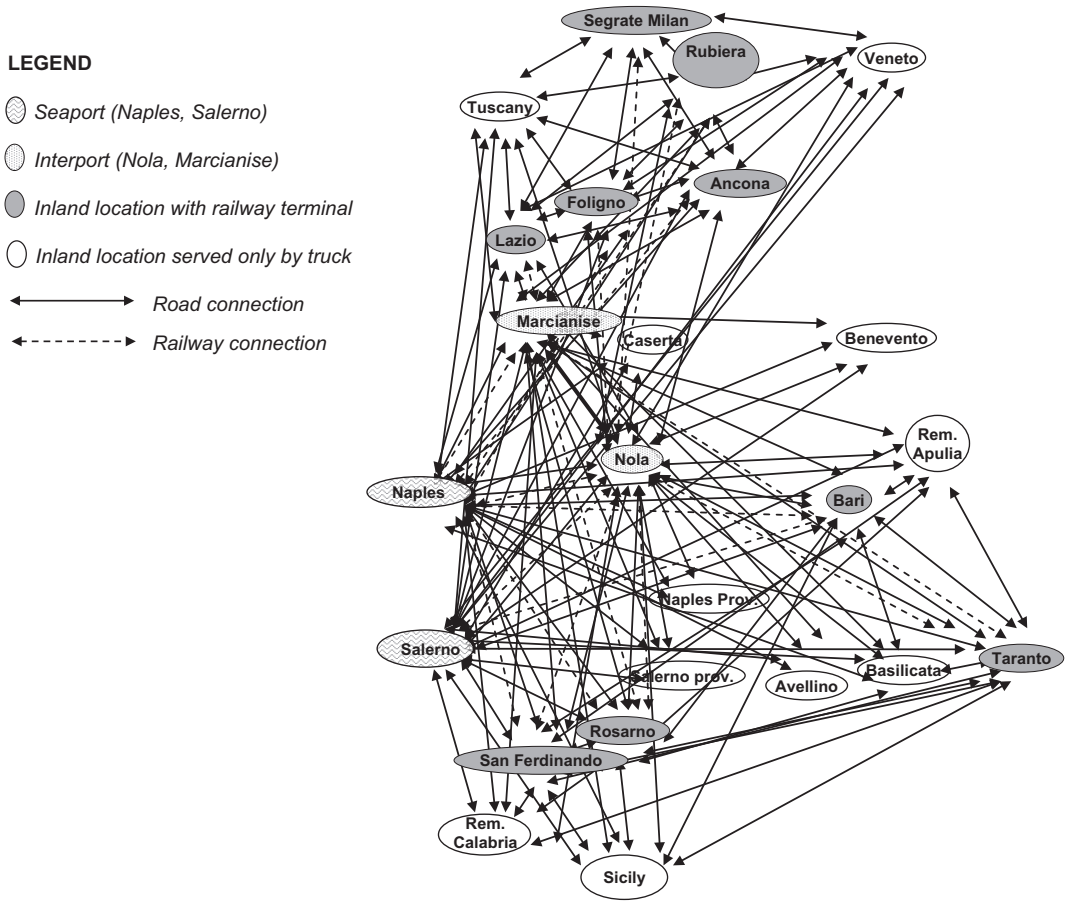


Fig. 2. The real port hinterland container network under consideration.

- *Inland locations accessible both by road and by rail*: Bari city/rail terminal; Taranto city/rail terminal, San Ferdinando city/rail terminal, Rosarno city/rail terminal, Lazio region/Civitavecchia rail terminal, Abruzzo and Marche region/Ancona rail terminal, Umbria region/Foligno rail terminal, Emilia Romagna region/Rubiera rail terminal, Lombardia region/Segrate Milan rail terminal.[18]
- *Inland locations accessible by road only*: Naples province, Salerno province, Caserta province, Avellino province, Benevento province, remaining part of Puglia region, Basilicata region, remaining part of Calabria region, Sicily region, Tuscany region, Veneto region.[19]

The two regional interports and the other twenty regional and extra-regional inland nodes above indicated represent the existing network of locations actually involved in the inland traffic of containers handled at the Campanian seaports.

The seaports of Naples and Salerno are connected by truck with all the inland locations of the network. Naples is also connected by railway with the regional interports of Nola and Marcianise and with some (but not all) extra-regional locations. Thus, the customs clearance of containers discharged and to be loaded at the port of Naples may take place either in the seaport itself or at the regional interports. Salerno is also connected by rail with the Nola interport and with the Bari city/terminal. There is no rail connection between Salerno and Marcianise. Hence, the customs clearance of containers discharged and to be loaded at the port of Salerno may take place either in the seaport itself or at the interport of Nola.

The interports are connected by road with all the other locations of the network, and by railway with some (but not all) extra-regional locations. The network under consideration also includes the following rail connections available in years preceding 2007: Naples-Foligno, Naples-Ancona, Nola-Foligno, Nola-Rubiera, Marcianise-Taranto and Marcianise-Civitavecchia.

[18] These locations are coded as: BAR, TAR, SAN, ROS, LAZ, ABR, UMB, EMI, MIL.
[19] These nodes are coded as: NAP2, SAL2, CAS, AVE, BEN, PUG, BAS, CAL, SIC, TOS, VEN.

All the inland nodes served by railway are assumed to also perform a multimodal transhipment function along various O/D traffic combinations.

Ultimately, each empirical model features 26 nodes and 219 admitted two way road and rail links (including two virtual nodes at each regional interport and their related connections), 18,071 unknowns and 1859 constraints. The two modelling applications have been programmed and solved with the *GAMS* (*General Algebraic Modeling System*) computer language, using the solver *CPLEX* (Brooke et al., 1998). The solutions have been obtained in negligible time on a Personal Computer with an Intel® Pentium® Dual CPU T3400 at 2.16 GHz and 2.87 GB of RAM, running under Microsoft® Windows® XP Professional. The non internalizing model has been solved in 0.53 s, while the model incorporating transport externalities has been solved in 0.59 s.

4.3. Main input data of empirical analysis

For the purpose of the analysis presented in this article, the framework of the input data and the structure of the port hinterland container distribution system suggested by Iannone (2012) are adopted. But differently from that earlier work, the objective function of the *interport model* now also covers the external costs of inland transport operations.

Main data were directly provided by various bodies and firms involved in intermodal logistics industry at regional, national and international level. Other data were derived from internet websites, specialized press, scientific literature, and industrial studies. Apart from the data concerning both a number of rail services (Salerno-Nola, Salerno-Bari, Naples-Ancona, Naples-Foligno, Nola-Foligno, Nola-Rubiera, Marcianise-Taranto, and Marcianise-Civitavecchia) and the unit external impacts of road and rail transport, all the input data used in the empirical models refer to the year 2007, which is a year particularly representative of the favourable economic and logistic conditions preceding those characterizing the ongoing global crisis.

A complete exposition of all input data is beyond the scope of this paper. In the present sub-section, some of the more significant data and their combinations are briefly commented. Most part of these data is shown in aggregated and/or average terms. More detailed information concerning can be obtained from the author upon request.

The aggregate supply of import and export full and empty containers at origin nodes and the aggregate O/D demands are reported in Table 1. Based on data diffused by the Campanian Port authorities, 750.457 full and empty TEU transited through the regional seaports in 2007 (excluding transhipment traffics), of which the 55% through the port of Naples. In particular, 332,576 full TEU and 77,657 empty TEU were imported and exported through Naples, while 232,515 full TEU and 107,709 empty TEU were imported and exported through the port of Salerno. O/D figures were estimated based on data and information provided by public and private companies operating in Campania. The hinterland of the Campanian seaport system

Table 1
Container supply and origin/destination demand at the Campanian seaports and at regional and extra-regional inland locations in 2007 (full and empty TEU). *Source:* author's processing based on data provided by the Campanian Port authorities, marine and inland terminal operators, transport and logistic companies.

Node	Import–export at the port of Naples (NAP)	Import–export at the port of Salerno (SAL)
NAP	*410,233*	*0*
SAL	*0*	*340,224*
NOL	74,424	2744
MAR	13,193	541
NAP2	178,220	56,994
SAL2	8416	127,101
CAS	22,264	16,473
AVE	10,469	7649
BEN	6503	12,855
PUG	15,457	22,361
TAR	4383	5727
BAR	17,294	7617
BAS	4470	8972
CAL	5730	615
ROS	6616	241
SAN	7081	221
SIC	636	1059
LAZ	26,184	56,364
ABR	4913	12,508
UMB	1917	33
TOS	636	17
EMI	765	101
MIL	26	13
VEN	636	18

Table 2
Probabilities of customs controls at the Campanian seaports in 2007. *Source:* author's processing based on data provided by the Italian Customs Agency, marine terminal operators, and freight forwarders.

Containerized imports	NAP (%)	SAL (%)
Automated computerized control – AC	59.0	84.0
Documentary control – DC	11.0	11.0
Physical inspection – PI	25.0	4.0
Scanner control – SC	5.0	1.0
Containerized exports		
Automated computerized control – AC	71.3	87.0
Documentary control – DC	10.0	10.0
Physical inspection – PI	15.6	2.5
Scanner control – SC	3.1	0.5

Table 3
Average dwell times of import and export containers at the Campanian seaports and interports in 2007 (number of days/TEU). *Source:* author's processing based on data provided by the Italian Customs Agency, marine and inland terminal operators, freight forwarders.

Dwell time of full containers	Days/TEU
Weighted average port dwell time of import containers cleared by customs at Naples and leaving by road/rail	14.0
Weighted average port dwell time of export containers entering by road/rail at Naples and cleared by customs	9.2
Weighted average port dwell time of import containers cleared by customs at Salerno and leaving by road/rail[a]	4.8
Weighted average port dwell time of export containers entering by road/rail at Salerno and cleared by customs[a]	4.7
Average port dwell time (at Naples and Salerno) for customs bonded containers transiting between regional seaports and interports by rail and under the responsibility of shipping lines[a]	1.7
Average interport dwell time of import/export containers cleared by customs at Nola and Marcianise and leaving/entering by road/rail[b]	6.1
Average interport dwell time at Nola and Marcianise for import/export containers cleared by customs at seaports and leaving/entering by road/rail[b]	4.6

[a] Availability of rail connections at the port of Salerno (modelling hypothesis).
[b] Availability of A3 area in the interport of Marcianise (modelling hypothesis).

encompasses areas of North, Centre and South of Italy. Anyway, the greatest volumes are originated and destined both in the Campania region itself and in Central and Southern Italy.

Table 2 reports data on the observed probabilities of customs controls at the Campanian seaports. The higher number of physical inspections and scanner examinations at Naples derives from elements contained in the customs declaration bills of the shippers and that are deemed potentially high-risky. Particularly for imports, at Naples a high number of customs controls takes place of containerized cargoes from China that directly compete with the Italian industry.

The deplorable conditions at the port of Naples – due to the under-dimensioning of customs facilities and staff, as well as to the lacking coordination of the various bodies in charge of technical and administrative controls – are notorious: delays in the release of containers up to several weeks are not uncommon. This leads to undesired increasing costs for shippers, including additional direct terminal operation costs and time related (indirect) costs. Table 3 lists data on average import/export dwell times at the Campanian container load centres. In Salerno there is not the same customs-induced port congestion as there is in Naples because it is a typical export port and imports arriving from China are very low.

Weighted average port figures were computed by taking account from one side the port dwell times by various types of customs control and leaving/entering transport mode, and from the other side the probabilities of customs controls reported in Table 2. Interport figures represent an average of dwell times by various types of customs control and leaving/entering transport mode. All port and interport dwell time figures employed are assumed to refer to the different possible cases of customs controls occurring in combination with other types of possible technical and administrative controls (veterinary controls, sanitary controls, etc.). The availability of rail connections at the Salerno port and the operational extended gateway status of the Marcianise interport are only hypotheses of the modelling applications. As a matter of fact, in 2007 there were neither rail connections at Salerno nor customs authorizations and facilities at Marcianise.

Table 4 lists the average costs of container release operations at the Campanian seaports and interports. These generalized cost figures include both direct costs of operations and time related indirect costs. Direct costs were computed by the formula: *[(Dwell time − demurrage free time) * demurrage charge] + terminal handling charge + [additional Customs costs (only for physically inspected and X-Ray scanned full containers)]*. Time related costs are the sum of in transit inventory holding costs and container leasing costs. Weighted average figures were computed by taking into consideration the probabilities observed in the Campanian seaports for the different customs control types (see Table 2).

Tables 5 and 6 lists the external cost and air emission parameters employed in the analysis of real life and modelled scenarios. These parameters have been calculated using specific marginal figures valid for Italy and released by the "Amici della

Table 4
Average total generalized logistic costs of container release operations at the Campanian seaports and interports in 2007 (Euros/TEU). *Source:* author's processing based on data provided by the Italian Customs Agency, marine and inland terminal operators, transport and logistic companies.

	NAP	SAL	NOL/NCC	MAR/MCC
Empty containers leaving/entering by road/rail[a]	504.6	172.2	111.1	111.1
Customs bonded full containers transiting between regional seaports and interports by rail and under the responsibility of shipping lines[a]	196.6	136.6	–	–
Weighted average for full containers cleared by customs in port and leaving/entering by road/rail[a]	626.5	204.8	–	–
Weighted average for full containers cleared by customs in interport and leaving/entering by road and rail (containers from/to Naples)[b]	–	–	178.5	178.5
Weighted average for full containers cleared by customs in interport and leaving/entering by road/rail (containers from/to Salerno)[a]	–	–	154.8	–
Full containers cleared by customs in ports and leaving/entering by road/rail	–	–	148.9	148.9

[a] Availability of rail connections at the port of Salerno (modelling hypothesis).
[b] Availability of A3 area in the interport of Marcianise (modelling hypothesis).

Table 5
Marginal external costs of container transport in Italy (Euros/TEU-km). *Source:* author's calculations based on data of Amici della Terra and Ferrovie dello Stato (2005).

	Road transport		Rail transport	
	Full containers	Empty containers	Full containers	Empty containers
Greenhouse gases	0.037	0.005	0.012	0.0017
Air pollution	0.142	0.020	0.020	0.0029
Noise	0.108	0.015	0.067	0.0095
Accidents	0.021	0.003	0.002	0.0002
Congestion	0.264	0.037	0.000	0.0000
Total	*0.572*	*0.081*	*0.101*	*0.014*

Terra'' association in collaboration with the Italian State Railway Company (Amici della Terra and Ferrovie dello Stato, 2005). Such figures were originally expressed in Euros/tonne-km and tonnes/tonne-km, and have been converted in Euros/TEU-km and Tonnes/TEU-km, respectively.[20] An average weight of containerized goods equalling 14 tonnes/TEU and a container tare of 2.3 tonnes/TEU have been assumed for these calculations. Based on data from Table 5, Fig. 3 demonstrates the total unit external benefits arising from shifting container movements from road to rail transport for different inland distance classes. Clearly, the longer the distance, the greater will be the total unit external cost saving from modal shift for any given transport volume. In general, transporting containers by train in Italy would produce about 6 times less total external cost per TEU-km than truck transport, which means that switching to rail would represent a reduction of slightly more than 82% in total negative externalities compared to the equivalent road journey.

Tables 7 and 8 are illustrative of the social generalized costs of transporting full containers between the two seaports, the two interports and just other three representative inland locations. Such figures are given by the sum of internal generalized transport cost and the external transport costs arising from greenhouse gas emissions, air and noise pollution, congestion and accidents. The dash (–) indicates that no direct connection exists or is admitted in the modelling applications.

4.4. Main results from empirical models and comparison with the real life situation

Table 9 gives evidence of the annual capacity utilization of the two-way rail container connections at the Campanian seaports and interports in both the observed real life situation and the modelled ideal scenarios. The underlying figures on rail traffic flows between the Campanian seaports and interports in the real situation also include possible transits of containers destined to and originating from extra-regional inland locations. The underlying figures on rail traffic flows between the Campanian interports and extra-regional locations in the real situation also include possible transits of containers not handled in the Campanian seaports but originating and terminating elsewhere. The underlying observed real figures on traffic between the Campanian seaports and extra-regional locations also include possible transits. The modelling solutions' figures exclusively refer to the capacity utilization due to shipments of containers imported and exported through the Campanian seaports, including transits.

Table 10 compares the railway transport's shares on the annual inland traffic of import and export containers handled at the seaports of Naples and Salerno in real life situation and modelled scenarios. Table 11 compares the annual rail traffic

[20] Amici della Terra and Ferrovie dello Stato (2005) did not estimate congestion costs associated to rail freight transport. Hence, as resulting from Table 5, it has not been possible to compute the marginal congestion costs of container transport by rail in Italy.

Table 6
Marginal physical air emissions of container transport in Italy (Tonnes/TEU-km). *Source:* author's calculations based on data of Amici della Terra and Ferrovie dello Stato (2005).

	Road transport		Rail transport	
	Full containers	Empty containers	Full containers	Empty containers
CO_2	0.00182560	0.00025760	0.00059006	0.00008326
$CO_2 + CH_4 + N_2O$ (CO_2 eq.)	0.00185519	0.00026178	0.00060729	0.00008569
SO_2	0.00000033	0.00000005	0.00000072	0.00000010
NO_x	0.00001528	0.00000216	0.00000158	0.00000022
PM_{10}	0.00000094	0.00000013	0.00000012	0.00000002
CO	0.00000470	0.00000066	0.00000050	0.00000007
VOC	0.00000232	0.00000033	0.00000020	0.00000003
Total greenhouse gases (CO_2 eq.)	*0.00185519*	*0.00026178*	*0.00060729*	*0.00008569*
Total air pollutants	*0.00002357*	*0.00000333*	*0.00000312*	*0.00000044*

CO_2, Carbon dioxide; CH_4, Methane gas; N_2O, Nitrous oxide; SO_2, Sulphur dioxide; NO_x, Nitrogen oxides; PM, Particulate Matter; CO, Carbon monoxide; VOC, Volatile Organic Compounds.

volumes from/to the interports of Nola and Marcianise for import/export containers transiting through the regional seaport cluster in the different cases of analysis. Table 11 also shows the rail transport's shares on the total annual inland traffic of import/export containers handled at the Campanian seaports and leaving/entering the interports. Table 12 lists the total internal, external and social logistic costs in the different cases. Table 13 compares the observed real situation and the modelled scenarios in terms of physical emissions of greenhouse gases and air pollutants deriving from inland container transport flows. Table 14 lists the optimal dual prices of rail capacity constraints.

It is worth pointing out that the overall utilization of rail capacity is greater in the modelled ideal policy scenarios than in the observed real life situation (Table 9). This is a remarkable result that mainly derives from the total internal generalized logistic cost reductions permitted by the employment of both the interports of Nola and Marcianise as extended gateways of the regional seaport system. The ideal scenario internalizing the externalities in objective function presents the greatest level of total two-way rail flows from and to the Campanian seaports and interports (96,146 TEU). Hence, this scenario features the greatest level of overall rail demand/supply balance (51.7%). More specifically, in the internalizing scenario, a volume of 88,964 TEU is shipped by rail from and to the Campanian seaports (also including the flows of containers destined to and arriving from the regional interports), which is approximately equivalent to a 141.9% increase from the real situation. The two way rail container flows between the Campanian interports and extra-regional inland locations in the same scenario amount to 7182 TEU.

On the whole, the rail flows between the Campanian interports and extra-regional inland locations generated by the ideal scenarios are lower than those characterizing the observed real situation (7728 TEU). This is easily explainable by considering the fact that, as already pointed out, the observed real figures on traffic between the regional interports and extra-regional locations resulting from Table 9 also incorporate transit flows of containers neither originating from nor destined to the seaports of Naples and Salerno. This category of flows has been not included in the empirical modelling applications presented in this paper. However, the full and empty containers discharged and loaded at the Campanian seaports and transiting by rail through the Campanian interports in 2007 amounted just to 142 TEU (Table 11).

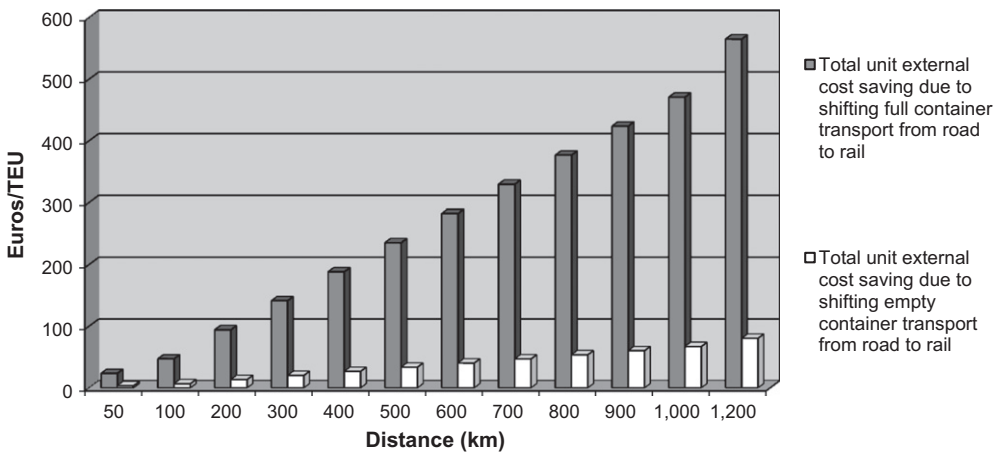


Fig. 3. Marginal external benefits from modal shift in container transport in Italy as a function of distance (Euros/TEU). *Source:* author's estimates based on data of Amici della Terra and Ferrovie dello Stato.

Table 7
Total social generalized costs for transport of full containers by road in 2007: representative locations in the Campanian port hinterland (Euros/TEU). *Source:* author's processing based on data provided by EAV-ANITA, Italian Customs Agency, Amici della Terra and Ferrovie dello Stato, transport and logistic companies, plus data derived from industrial studies, internet web-sites and specialized press.

From/to	NAP	SAL	NOL/NCC (import)	NOL/NCC (export)	MAR/MCC (import)	MAR/MCC (export)	ROS	LAZ	MIL
NAP	–	–	198.7	–	198.7	–	901.7	486.6	1525.2
SAL	–	–	283.2	–	296.6	–	808.4	578.5	1605.8
NOL/NCC (import)	–	–	–	–	138.7	–	890.2	480.4	1523.5
NOL/NCC (export)	198.8	283.5	–	–	–	138.8	–	–	–
MAR/MCC (import)	–	–	138.7	–	–	–	936.4	428.8	1480.0
MAR/MCC (export)	198.8	296.9	–	138.8	–	–	–	–	–

Table 8
Total social generalized costs for transport of full containers by rail in 2007: representative locations in the Campanian port hinterland (Euros/TEU). *Source:* author's processing based on data provided by railway companies, Italian Customs Agency, Amici della Terra and Ferrovie dello Stato, plus data derived from industrial studies and specialized press.

From/to	NAP	SAL	NOL/NCC (import)	NOL/NCC (export)	MAR/MCC (import)	MAR/MCC (export)	ROS	LAZ	MIL
NAP	–	–	168.7	–	166.2	–	300.8	–	–
SAL	–	–	208.3	–	–	–	–	–	–
NOL/NCC (import)	–	–	–	–	–	–	454.8	–	597.2
NOL/NCC (export)	169.7	209.6	–	–	–	–	–	–	–
MAR/MCC (import)	–	–	–	–	–	–	445.8	337.6	–
MAR/MCC (export)	167.3	–	–	–	–	–	–	–	–

Table 9
Observed real data and modelled ideal scenario's data on capacity utilization of rail container connections from/to the Campanian seaports and interports. *Source:* author's processing on data provided by railway companies and inland terminal operators, plus data resulting from modelling applications.

Two-way rail services	Annual maximum capacity during 2005–2007 (TEU)	Capacity utilization in the last observed year during 2005–2007 (%)	Annual capacity utilization resulting from the internalizing ideal scenario (%)	Annual capacity utilization resulting from the non internalizing ideal scenario (%)
NAP-NOL/NCC	30,000	30.9	100.0	100.0
NAP-MAR/MCC	6000	8.0	100.0	100.0
NAP-BAR	25,000	49.2	69.2	69.2
NAP-ROS	15,000	31.0	96.4	96.4
NAP-SAN	5000	102.6	100.0	100.0
NAP-ABR	5000	1.8	72.3	72.3
NAP-UMB	5000	1.4	51.1	51.1
NAP-EMI	5000	2.6	19.1	28.5
NOL/NCC-TAR	15,000	21.5	6.9	9.1
NOL/NCC-ROS	10,000	18.8	0.0	7.8
NOL/NCC-SAN	24,000	3.7	0.0	3.3
NOL/NCC-UMB	5000	1.0	0.0	0.3
NOL/NCC-EMI	5000	1.3	0.0	0.0
NOL/NCC-MIL	6000	25.0	9.8	2.0
MAR/MCC-TAR	5000	1.5	0.0	50.0
MAR/MCC-ROS	5000	0.2	0.0	0.0
MAR/MCC-LAZ	5000	0.8	32.1	32.1
SAL-NCC/NOL	5000	8.8	54.2	72.2
SAL-BAR	5000	84.2	100.0	100.0
TOTAL	186,000	23.9	48.8	51.7

In light of the above results, the market shares of railway transport on the annual inland traffic of containers discharged and loaded in the seaports of Naples and Salerno result bigger in the simulated ideal scenarios than in the real situation (Table 10). As far as the two-way inland container traffic between the regional seaports and interports is particularly concerned, the ideal scenario internalizing the externalities in objective function presents a lower market share of rail transport than that obtained from the solution of the non internalizing ideal scenario. This result may in the first instance seem odd but can be straightforwardly interpreted by considering both the capacity saturation of the rail connections between seaports and interports under both simulated scenarios (Table 9) and the fact that, under the internalizing scenario, most of the containers carried by road between the seaports and the interports result to be only in transit through the interports. In particular, the social marginal cost charging of inland transport operations favours a greater employment of the rail services between interports and extra-regional locations for sending and receiving the import and export containers discharged

Table 10
Market share of railway transport in inland container traffic from/to the Campanian seaports (real situation and modelled ideal scenarios).

Market share of railway transport		From/to Naples (%)	From/to Salerno (%)	From/to Naples and Salerno (%)
Total inland traffic	Observed real situation	7.8	0.0	4.3
	Non internalizing ideal scenario	19.5	2.3	11.7
	Internalizing ideal scenario	19.6	2.5	11.9
Inland traffic to/from the Campanian interports	Observed real situation	10.4	0.0	10.1
	Non internalizing ideal scenario	38.4	79.7	39.9
	Internalizing ideal scenario	37.1	82.0	39.0
Inland traffic to/from other inland locations	Observed real situation	7.0	0.0	3.4
	Non internalizing ideal scenario	13.9	1.5	7.5
	Internalizing ideal scenario	14.2	1.5	7.6

Table 11
Market share of railway transport in inland traffic from/to the Campanian interports for containers discharged/loaded at the Campanian seaports (real situation and modelled ideal scenarios).

	Observed real situation	Non internalizing ideal scenario	Internalizing ideal scenario
Containers discharged/loaded at Naples			
Railway traffic leaving/entering Nola and Marcianise (full and empty TEU)	142	3108	6065
Railway transport's share of total inland traffic leaving/entering Nola and Marcianise	2.2%	9.5%	18.5%
Containers discharged/loaded at Salerno			
Railway traffic leaving/entering Nola and Marcianise (full and empty TEU)	0	114	1117
Railway transport's share of total inland traffic leaving/entering Nola and Marcianise	0.0%	100.0%	100.0%
Containers discharged/loaded at Naples and Salerno			
Railway traffic leaving/entering Nola and Marcianise (full and empty TEU)	142	3222	7182
Railway transport's share of total inland traffic leaving/entering Nola and Marcianise	2.2%	9.8%	21.2%

and to be loaded at the Campanian seaports (Tables 9 and 11). Hence, the rail transport's share of total inland traffic between regional interports and extra-regional nodes in the internalizing scenario is greater than that resulting from the optimal solution of the non internalizing scenario (Table 11).

The possibility of arranging the transfer of containers by railway carrier haulage under customs bond between seaports and interports dramatically reduces the generalized cost of release operations at the seaports, favouring a full utilization of the rail capacity connecting seaports and extended gateways. In this respect, it seems worth to point out that it is frequently argued that rail transport can only compete with road transport over long distances. Generally, a distance of 250–300 km represents the shortest viable distance for rail to adequately compete with road services. The results of the modelled scenarios demonstrates that this argument is extremely simplistic, since other factors which may relate, for instance, to the saturation of terminal capacity in seaports as well as to regulation (i.e. the trade facilitation permitted by the possibility of using the interports as extended gateways) may favour rail over road below this range. To sum up, under a regime of customs continuity, rail transport may offer a very competitive alternative to road transport on the short distance routes connecting the Campanian seaports and interports. In addition, the adoption of the extended gateway practice, backed up by an efficient railway system, could prove to be an effective policy measure to expand the hinterland reach of the Campanian seaport cluster.

Nevertheless, shippers failed to seize even the customs clearance opportunities arisen at Nola in the past. In 2007, their behaviour was far from being optimal, and the strategic two-way rail connection between Naples and Nola was used at less than a third of its full capacity (Table 9). Such a finding leads to confirm that, in the field of intermodal logistics, a number of historical, psychological, political and personal factors, also including opportunistic behaviour, inertia and bounded rationality of economic agents, can often result in practical choices actually diverging from the theoretical optimum (Notteboom, 2001).

Table 12
Internal, external and social logistic costs for inland distribution of full and empty containers imported and exported through the Campanian seaports (real situation and modelled ideal scenarios).

	Observed real situation	Non internalizing ideal scenario	Internalizing ideal scenario
Total internal generalized logistic cost for inland distribution of containers discharged and loaded in Naples (millions of Euros)	312.68	298.55	298.61
Total internal generalized logistic cost for inland distribution of containers discharged and loaded in Salerno (millions of Euros)	128.42	128.13	128.26
Total internal generalized logistic cost for inland distribution of containers discharged and loaded in Naples and Salerno (millions of Euros)	*441.09*	*426.69*	*426.87*
Total external cost of greenhouse gas emissions from road and railway transport of containers discharged and loaded in Naples and Salerno (millions of Euros)	2.25	2.02	1.98
Total external cost of air pollution from road and railway transport of containers discharged and loaded in Naples and Salerno (millions of Euros)	8.36	7.25	7.05
Total external cost of noise pollution from road and railway transport of containers discharged and loaded in Naples and Salerno (millions of Euros)	6.64	6.36	6.32
Total external cost of accidents from road and railway transport of full and empty containers discharged and loaded in Naples and Salerno (millions of Euros)	1.24	1.06	1.03
Total external cost of congestion from road transport of containers discharged and loaded in Naples and Salerno (millions of Euros)	15.35	12.87	12.43
Global total external cost for road and railway transport of containers discharged and loaded in Naples and Salerno (millions of Euros)	*33.84*	*29.56*	*28.82*
Total social generalized logistic cost for inland distribution of containers discharged and loaded in Naples and Salerno (millions of Euros)	***474.93***	***456.24***	***455.69***

Table 13
Tonnes of atmospheric **emissions from inland** transport of containers imported and exported through the Campanian seaports (real situation and modelled ideal scenarios).

	Observed real situation	Non internalizing ideal scenario	Internalizing ideal scenario
CO_2	109500.3	98763.4	96946.0
Total greenhouse gases (CO_2 eq.)	*111319.4*	*100492.0*	*98660.3*
CO	276.0	237.3	230.6
NO_x	897.3	770.8	748.9
PM	55.1	47.8	46.5
SO_2	23.3	28.1	29.0
VOC	136.2	116.4	113.0
Total air pollutants	*1387.9*	*1200.3*	*1168.0*

Table 14
Shadow prices of the capacity limits of rail connections for the inland distribution of full and empty containers imported and exported through the Campanian seaports (Euros/TEU).

Two-way rail services	Shadow price of the capacity limit for import containers (non internalizing ideal scenario)	Shadow price of the capacity limit for import containers (internalizing ideal scenario)	Shadow price of the capacity limit for export containers (non internalizing ideal scenario)	Shadow price of the capacity limit for export containers (internalizing ideal scenario)
NAP-NOL/NCC	375.9	388.7	4.8	9.3
NAP-MAR/MCC	375.9	391.2	17.3	114.7
NAP-BAR	0.0	0.0	0.0	0.0
NAP-ROS	313.3	334.7	0.0	0.0
NAP-SAN	331.0	357.3	14.1	14.8
NAP-ABR	0.0	0.0	35.2	154.5
NAP-UMB	0.0	0.0	0.0	0.0
NAP-EMI	0.0	0.0	0.0	0.0
NOL/NCC-TAR	0.0	0.0	0.0	0.0
NOL/NCC-ROS	0.0	0.0	0.0	0.0
NOL/NCC-SAN	0.0	0.0	0.0	0.0
NOL/NCC-UMB	0.0	0.0	0.0	0.0
NOL/NCC-EMI	0.0	0.0	0.0	0.0
NOL/NCC-MIL	0.0	0.0	0.0	0.0
MAR/MCC-TAR	0.0	3.9	0.0	0.0
MAR/MCC-ROS	0.0	0.0	0.0	0.0
MAR/MCC-LAZ	0.0	0.0	0.0	0.0
SAL-NCC/NOL	43.1	46.5	0.0	0.0
SAL-BAR	36.0	51.4	35.8	145.1

As for total costs (Table 12), the ideal policy scenarios present lower levels of internal and external logistic costs than those observed in the real life situation. The total internal cost saving resulting from the optimized scenarios ranges between 3.22% and 3.27%, whereas the total external cost saving ranges between 12.64% and 14.82%.

When comparing only the two simulated ideal scenarios, the trade-off between total internal costs and total external costs is obvious. The internalization of transport external costs in the objective function stimulates rail-based container distribution solutions, thus determining slightly higher internal generalized logistic costs and decisively lower external costs than those resulting from the non internalizing scenario. The total internal generalized cost for inland distribution of full and empty containers discharged and loaded in the Campanian seaports would amount to €426.69 million and €426.87 million in non internalizing scenario and in internalizing scenario, respectively. The total external cost in the two different case studies would amount to €29.56 million and €28.82 million. Thus, the scenario incorporating externalities in objective function would feature the lowest total social generalized inland container logistics cost. Compared with the real life situation, the overall cost reduction is slightly more than 4%, amounting to €19.24 million.

Compared with the existing situation, the optimized flows in simulated ideal scenarios would also generate a total saving of greenhouse gas emissions ranging between 9.7% and 11.4%, whereas the total reduction of air pollutants would range between 13.5% and 15.8% (Table 13). The scenario incorporating externalities in objective function features the lowest total atmospheric emissions. In particular, it has been estimated an annual saving of the order of about 12,660 tonnes of CO_2 equivalent emissions from transport corresponding to an external cost reduction of 0.27 million euros from the observed real life situation, whereas the estimated saving in terms of air pollution (CO, NO_x, PM, SO_2, VOC) from transport is approximately 220 tonnes per year corresponding to an external benefit of 1.31 million euros.

The computer solutions of the modelling applications have also provided with marginal values corresponding to the optimal dual variables associated to primal constraints. Table 14 shows the shadow prices corresponding to the capacity limits of rail connections in both simulated ideal policy scenarios. These prices indicate the imputed value of the objective function that would arise from an improvement of rail infrastructure and/or services.

For instance, based on the results obtained from the internalizing scenario it would be possible to obtain internal and external benefits from the improvement of the two-way rail connections of Naples-Nola (NAP-NOL/NCC), Naples-Marcianise (NAP-MAR/MCC), Naples-San Ferdinando (NAP-SAN), Salerno-Bari (SAL-BAR). In addition, also the improvement of the one-way connections of Naples-Rosarno (NAP-ROS), Ancona-Naples (ABR-NAP), Marcianise-Taranto (MAR/MCC-TAR) and Salerno-Nola (SAL-NOL/NCC) would generate reductions of the total social generalized logistic cost under the same scenario.

Due to the social marginal cost charging of transport operations, the shadow prices resulting from the internalizing scenario are higher than those characterizing the optimal solution of the other modelled scenario. For instance, the total social generalized logistic cost reduction deriving from a unit relaxation of the capacity constraint of the railway connection from Salerno to Nola (SAL-NOL/NCC) would equal €46.5 under the internalizing scenario, whereas the total private generalized logistic cost reduction arising from a unit relaxation of the same capacity limit would equal €43.1 under the non internalizing scenario. In general, the optimal dual prices of all port-interport rail connections' capacity limits under both simulated ideal policy scenarios would confirm the importance of a fully operational extended gateway role of both regional interports to support the inland traffic of import and export containers transiting through the regional seaport system.

Presumably, also other rail connections not indicated above would result more and even fully utilized under a possible scenario of increased inland O/D demand of containers discharged and to be loaded at the Campanian seaports. Actually, the port of Naples is currently expanding its container handling capacity by 1 million TEU in anticipation of expected future increases in traffic volumes. As a consequence, an even greater level of rail connections' capacity than that assumed in the simulated policy scenarios described so far would be needed to meet rising trade volumes and match the additional port capacity. Without such an improvement of rail connections, the shadow prices of capacity limits would result much higher than those obtained from the simulated scenarios discussed above. However, the scenarios presented in this article conservatively assume the container demand volumes observed in 2007.

4.5. Sensitivity analysis

Over the long term road transport costs are expected to increase due to rising oil prices, congestion, drivers' hours regulations and other factors that are likely to have a negative impact on truck freight flows and might encourage a shift to other transport solutions. Considering the high degree of uncertainty surrounding long term projections, a sensitivity analysis based on the *interport model* has been executed to inform about possible future port hinterland container flows under different scenarios concerning a number of hypothetical developments in road transport fares.

Fig. 4 presents the impact of road transport prices' variations on rail transport's market share in the modal split of total optimal inland flows of containers leaving and entering the Campanian seaports and interports under the ideal policy scenario without externalities in objective function. Compared with the base case, a range of road fares' variations are analyzed: (i) 90% reduction; (ii) 75% reduction; (iii) 50% reduction; (iv) 25% reduction; (v) 25% increase; (vi) 50% increase; (vii) 75% increase; (viii) 90% increase. As the figure illustrates, the maximum market share of rail transport which can be achieved is approximately constant under road transport prices' variations ranging between 50% and 90%.

In order to compare the effects of road fares' variations with the effect of the internalization of external costs, Table 15 presents the market share of rail transport at the Campanian seaports and interports under the different modelled scenarios and the real life situation. It emerges that the effects of road fares' growth under the modelled scenario without externalities

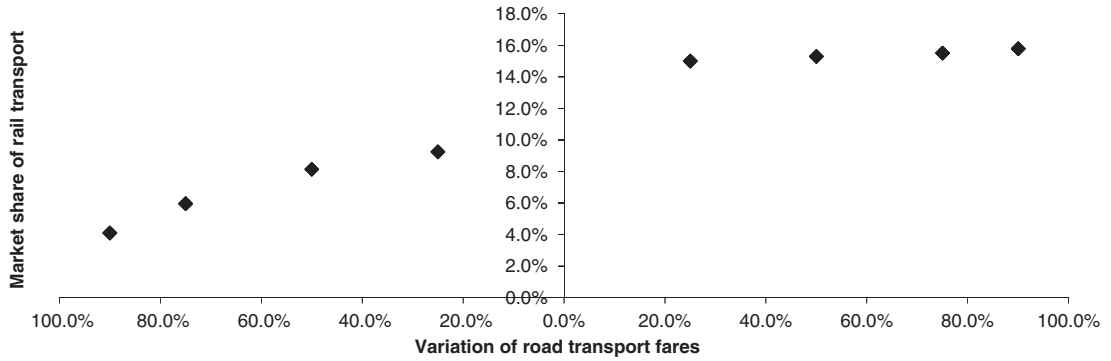


Fig. 4. Rail transport's share in modal split of total inland container flows from and to the Campanian seaports and interports as a function of road transport fares' variation (non internalizing modelled scenario).

Table 15
Road transport fares' variations and internalization of transport external costs: effects on railway transport's share in modal split of total inland container flows at the Campanian seaports and interports.

Market share of rail transport (observed real situation in 2007) (%)	Market share of rail transport (non internalizing ideal scenario – BASE case) (%)	% Variation of road transport fares (non internalizing ideal scenario – BASE case's variations) (%)	Market share of rail transport (as a result of the variation of road transport fares; non internalizing ideal scenario – BASE case's variations) (%)	Market share of rail transport (internalizing ideal scenario) (%)
4.9	14.3	−90.0	4.1	14.9
		−75.0	6.0	
		−50.0	8.1	
		−25.0	9.2	
		25.0	15.0	
		50.0	15.3	
		75.0	15.5	
		90.0	15.8	

(see the fourth column of the table) are increasingly larger than that obtained by stimulating modal shift through a social marginal cost charging policy (see the fifth column of the table). However, in any case it can be argued that promoting a shift towards sustainable inland container transport will help minimize exposure to high transport costs and limit external diseconomies.

5. Conclusions

Much attention is currently being given to seamless intermodal distribution solutions in relation to the optimization of port hinterland container logistics. On the other hand, hinterland container connections have become a crucial element for the competitiveness both of seaports and of the wider global and regional productive systems the seaports belong to, affecting the internal and external costs of production–distribution and logistic supply chains.

This article has investigated a number of conditions to achieve private and social cost efficiency in the landside distribution of maritime containers transiting through the seaports and interports of the Campania region in Southern Italy. The advantages arising from improved rail connections and from the practical implementation of the extended gateway concept have been empirically demonstrated. In addition, the role of internalization of transport external costs has been put into evidence.

Port hinterland container logistics is a network industry in which it is involved a wide range of public and private stakeholders and that is made up of several components such as infrastructure systems, equipments, transport vehicles, and various types of regulations and procedures. The performance of a port hinterland container distribution network system depends on the quality of interactions between its participating actors and on the optimal functioning of all its constituent elements in combination. A lack of clear and effective policies and institutional arrangements, or competing interests among actors, can pose severe threats to the efficiency and growth of the system.

In order to achieve greater competitiveness and sustainability of port hinterland container distribution operations through the Campanian load centers, it is essential to think holistically about the provision, regulation and use of regional container infrastructures and services. The case studies illustrated in this paper reinforce the importance of adopting an integrated package of public and private policies to simultaneously achieve economic, environmental and social benefits.

A full integration of container operations between the regional seaports and interports can relieve seaport congestion phenomena and promote the shift of standardized loading units from road-only transport to intermodal rail transport. How-

ever, the availability of customs facilitations at the interports will be of little help as long as the railway connections between seaports and interports remain inadequate. This has been essentially demonstrated by the full utilization of these connections achieved as a result of the modelling applications. Moreover, this also implies the removal of rail infrastructure bottlenecks (particularly at the port of Salerno) as well as the introduction of a more effective policy approach to the liberalization and privatization of rail traction services between seaports and interports. Both technical and bureaucratic barriers need therefore to be lifted urgently. Private rail companies are not still eligible for providing transit services under facilitated conditions between seaports and interports. At the moment, only the incumbent is not obliged by law to stipulate expensive surety policies for guaranteeing the cargoes during the transportation under customs bond on behalf of shipping lines.

The results obtained from the simulated scenarios can be also used as a reference for public policy makers of the region to adopt rational decisions on the internalization of transport externalities in the aim of supporting sustainable transport solutions for port hinterland container logistics. As demonstrated in this paper, the internalization of external diseconomies in transport prices can lead to a greater use of railway services between the regional interports and extra-regional locations, thus suggesting as a possible adequate policy measure that of providing a grant scheme for funding container shuttle trains on these routes based on the total external cost savings deriving from shifting containers from unimodal road transport. By introducing such a type of incentives, it would be possible to implicitly implement a well suit pricing system that takes care of a better spread of traffic and a more fair division between the different modes of transport connecting the interports of Nola and Marcianise to other locations outside the Campania region for the forwarding of import and export containers transiting through the regional seaports. This policy measure could resolve, at least in part, the current bottlenecks in some connections of the seaports of Naples and Salerno with their distant hinterland.

Possible future applications of the *interport model* could concern the evaluation of policy scenarios for other container network systems in Italy and Europe, while also including barge transport and taking account of a number of recently introduced industrial practices such as the cross-border extended gateway logistics and the port hinterland container distribution arranged by maritime terminal operators. The *interport model* can be a also useful tool to evaluate possible changes in hinterland contestability arising from different policy scenarios concerning the sea-land intermodal logistics system of multiple regions in a country, as well as that of an entire country or even a group of countries. An example could be the so called "ARA region" (Amsterdam–Rotterdam–Antwerp) in Northern Europe.

Acknowledgements

Early preliminary results of this study were presented at the European Transport Conference (ETC) 2011 organized by the Association of European Transport (AET) from 10 to 12 October 2011 in Glasgow (Scotland, UK). The delivery of those results was made possible thanks to the financial research grants awarded to the author by the Institute for Service Industry Research (IRAT) of the National Research Council of Italy (CNR).

The suggestions of the journal's editor and the advices of three anonymous referees have been extremely helpful in improving the research outcomes discussed in the present article. The Department of Economics and Quantitative Methods (DIEM) at the Faculty of Economics of the University of Genoa in Italy is acknowledged for currently supporting the research activities of the author. However, the author is solely responsible for the content presented in this article as well as for possible errors and omissions.

Appendix A. Supplementary material

Supplementary material associated with this article can be found, in the online version, at http://dx.doi.org/10.1016/j.tra.2012.05.019.

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